

NEHALA MOHAMED

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PROFILE SUMMARY

Machine Learning enthusiast with practical experience in building predictive models, deep learning systems, and NLP applications. Proficient in data preprocessing, feature engineering, and exploratory data analysis (EDA). Skilled in Python, TensorFlow, and Scikit-Learn, with a strong interest in Generative AI, including LLMs and transformers. A proactive self-learner who thrives on continuous learning and applying cutting-edge AI techniques to solve real-world challenges and deliver business impact.

TECHNICAL SKILLS

- **Programming:** Python, Django, SQL, Flask
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn
- **Databases:** PostgreSQL
- **Machine Learning:** Linear Regression, Random Forest, KNN, Naive Bayes, SVM, Decision Trees, XGBoost, K-Means, DBSCAN
- **Advanced Topics:** NLP, Deep Learning, Transformers, LLMs, GenAI, CNN
- **Cloud:** AWS (EC2), Docker, Azure ML
- **MLOps:** CI/CD, GitHub, Docker, Model Deployment
- **Big Data Technologies:** Hadoop, Spark, Hive, Sqoop
- **Agentic AI Frameworks:** LangChain, RAG (Retrieval-Augmented Generation), Streamlit

PROJECTS

Heart Disease Prediction Using Machine Learning (Classification & Regression) *Python, Scikit-learn, Pandas, Matplotlib, Seaborn*

- Conducted a comprehensive analysis on a heart disease dataset to predict disease presence (classification) and estimate severity/risk score (regression), applied preprocessing techniques, Built and compared multiple classification models (Random Forest Classifier) with metrics.

Time Series Forecasting of Milk Production *Python, Pandas, ARIMA, Prophet, LSTM, Exponential Smoothing*

- Developed time series forecasting models to predict monthly milk production using historical dairy industry data, implemented and compared forecasting models: **ARIMA**, **Exponential Smoothing**, **Facebook Prophet**, and **LSTM Neural Networks**.

NLP Project: Public Sentiment Analysis on COVID-19 Vaccination *Python, NLTK, Scikit-learn, TensorFlow, LSTM, Word2Vec*

- Collected and preprocessed text data from social media platforms, Performed text cleaning and feature extraction using TF-IDF and Word2Vec embeddings, implemented sentiment analysis models using **Logistic Regression**, **Support Vector Machines (SVM)**, and **LSTM neural networks**. Used Named Entity Recognition (NER)

Image Classification using CNN

Python, TensorFlow, Keras, OpenCV, CNN

Image Classification using CNN (CIFAR-10) – Developed and trained a deep Convolutional Neural Network (CNN) model on the CIFAR-10 dataset, achieving over 88% validation accuracy. Implemented preprocessing, data augmentation, and optimization using Adam optimizer with categorical cross-entropy loss. Evaluated performance using accuracy/loss visualization, confusion matrix, and real-world testing.

Deep Learning Project: Image Generation using GAN

Python, TensorFlow, Keras, GAN, NumPy, Matplotlib

Image Generation using GAN (MNIST) – Implemented a Generative Adversarial Network (GAN) with Generator and Discriminator models trained adversarially on the MNIST dataset. Fine-tuned hyperparameters and stabilized training using binary cross-entropy loss. Achieved realistic image generation after 10,000+ iterations, demonstrating effective learning of image features.

Healthcare Data Analysis using Hive

Hive, Google Colab, Pandas, Matplotlib, Python, Sqoop

Healthcare Data Analysis using Hive in Google Colab – Analyzed patient datasets containing demographics, diseases, and hospital information to uncover key healthcare trends. Performed data preprocessing, EDA, and executed Hive queries with Sqoop simulation for scalable analytics. Derived actionable insights on disease patterns, hospital workloads, and city-level health distributions.

Findora AI Search Engine (Chatbot + Voice Output)

Python, Streamlit, LangChain, Llama 3, SpeechRecognition, pyttsx3

AI-Powered Search Engine and Chatbot using LangChain & Llama 3 – Developed an interactive Streamlit-based chatbot integrated with LangChain for contextual query handling and memory management. Utilized the Llama 3 model via Groq API for efficient natural language generation and added voice output using pyttsx3. Demonstrated Agentic AI principles by combining perception, reasoning, and action in real-time interaction.

Findora AI – RAG-powered Intelligent Chat System

Python, FastAPI, PostgreSQL, LangChain, Groq, Redis, Celery, Streamlit

- Developed a secure RAG-based AI platform enabling PDF knowledge extraction and conversational search with LangChain-driven sparse and dense retrieval pipelines. Integrated Groq-powered LLaMA models with vision support, real-time WebSocket communication, Redis caching, Celery background task processing, and external web search. Implemented JWT authentication, RESTful API best practices with proper HTTP responses, CI/CD pipeline automation, and an interactive Streamlit UI with Swagger API documentation.

EDUCATION

Machine learning Brototype, Remote	Jul 2024 – Present
Python Django Avodha	jan 2023 – may 2023
Btech in Computer Science Royal College of Engineering	aug 2014 –jun 2018 CGPA: 7.00